Example 1.32

Is 400 a perfect cube?

Solution:

By prime factorisation, we have $400 = \underline{2 \times 2 \times 2} \times 2 \times 5 \times 5$.

There is only one triplet. To make further triplets, we will need two more 2's and one more 5. Therefore, 400 is not a perfect cube.

Example 1.33

Find the smallest number by which 675 must be multiplied to obtain a perfect cube.

Solution:

We find that, $675 = 3 \times 3 \times 3 \times 5 \times 5$ .....(1)

Grouping the prime factors of 675 as triplets, we are left over with 5×5.

We need one more 5 to make it a perfect cube.

To make 675 a perfect cube, multiply both sides of (1) by 5.

$675 \times 5 = 3 \times 3 \times 3 \times 5 \times 5 \times 5$

$3375 = 3 \times 3 \times 3 \times 5 \times 5 \times 5$

Now, 3375 is a perfect cube. Thus, the smallest required number to multiply 675 such that the new number perfect cube is 5.

3	675
3	225
3	75
5	25
5	1
	1



Think

In this question, if the word 'multiplied' is replaced by the word 'divided', how will the solution vary?

1.8.3 Cube root of a given number by Prime Factorisation

Step 1: Resolve the given number into the product of prime factors.

Step 2: Make triplet groups of same primes.

Step 3: Choosing one from each triplet, find the product of primes to get the cube root.

Example 1.34

Find the cube root of 27000.

Solution:

By prime factorisation, we have $27000 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 5 \times 5 \times 5$

$$\therefore \sqrt[3]{27000} = 2 \times 3 \times 5 = 30.$$

Example 1.35

Evaluate: (i) $\sqrt[3]{\frac{9261}{8000}}$ (ii) $\sqrt[3]{\frac{1728}{729}}$

Solution:

(i) $\sqrt[3]{\frac{9261}{8000}} = \frac{\sqrt[3]{9261}}{\sqrt[3]{8000}} = \frac{(21 \times 21 \times 21)^{\frac{1}{3}}}{(20 \times 20 \times 20)^{\frac{1}{3}}} = \frac{\left(21^3\right)^{\frac{1}{3}}}{\left(20^3\right)^{\frac{1}{3}}} = \frac{21}{20} = 1\frac{1}{20}$

(ii) $\sqrt[3]{\frac{1728}{729}} = \frac{\sqrt[3]{1728}}{\sqrt[3]{729}} = \frac{(12 \times 12 \times 12)^{\frac{1}{3}}}{(9 \times 9 \times 9)^{\frac{1}{3}}} = \frac{\left(12^3\right)^{\frac{1}{3}}}{\left(9^3\right)^{\frac{1}{3}}} = \frac{12}{9} = \frac{4}{3} = 1\frac{1}{3}$

01_8th_Maths_Numbers_Chap 1.indd 38 2/12/2020 12:05:12 PM